

Is the AI Any Good?

Why evaluating AI is hard · Three ways to check answers · ROUGE, BLEU and BERTScore · Bedrock model evaluations · Responsible AI

Why evaluating AI is hard

Normal software	Generative AI
2 + 2 always returns 4	The same question can get different answers
One right answer	Many good answers, worded differently
A test passes or fails	"Better" is often a judgement



"Summarise this complaint."

Two summaries can use completely different words — and both be good.

Three ways to check answers

Method	How it works	Good for
People	Humans read and rate answers	Tone, helpfulness, judgement calls
Scores	Compare to a reference answer automatically	Large volumes, quick checks
LLM as judge	Another model grades the answer	Scale and some judgement

 **Most teams use all three:** scores for speed, a judge model for scale, and people for the cases that matter most.

ROUGE, BLEU and BERTScore

Score	What it checks	Typical use
ROUGE	How many of the reference's words appear in the answer	Summaries
BLEU	How many of the answer's words appear in the reference	Translation
BERTScore	Whether the meaning matches, even with different words	Paraphrased answers



Reference: "The parcel arrived late and damaged." **Answer:** "Delayed delivery, broken item."
ROUGE and BLEU score this **low** — few shared words. BERTScore scores it **high** — same meaning.

Amazon Bedrock model evaluations

Bedrock can run evaluations for you.

Type	What happens
Automatic	Scores accuracy, robustness and toxicity on a dataset
Human	Your own team rates the answers
LLM as judge	A judge model rates correctness, helpfulness and more



Use a built-in dataset — or your own questions and reference answers.



Your own questions test your use case. A general dataset doesn't.

Checking before you release

Never switch to a new model or prompt without checking it first.

Step	What you do
1 Golden set	Keep real questions with agreed good answers
2 Compare	Run the new version against it and compare scores
3 Few users first	Release to a small group, then everyone
4 Agent tracing	See each step an agent took, and why



A new prompt scores higher overall — but the trace shows it skips the refund-policy lookup. Caught before customers see it.

Responsible AI

AWS describes responsible AI in eight areas.

Principle	In plain words	Principle	In plain words
Fairness	Treats groups of people equally	Controllability	People can monitor and steer it
Explainability	You can see why it answered	Veracity & robustness	Accurate, even with odd input
Privacy & security	Data is protected	Governance	Clear rules and owners
Safety	Avoids harmful output	Transparency	People know it's AI



You can't claim a system is fair or accurate unless you've measured it.

Key takeaways

- 1 AI answers can't be checked with a simple pass or fail
- 2 Use people, scores and a judge model together
- 3 ROUGE and BLEU count words — BERTScore checks meaning
- 4 Test with your own questions, not just general ones
- 5 Check every change against a golden set before release

Next · Part 2: watching your AI in production